Fahad Nadim Ziad

Dhaka, Bangladesh • +8801676212004 • f.n.ziad@gmail.com
ziaaad.vercel.app   linkedin.com/in/fahadnadimziad   github.com/fnziad

Education

BRAC University — Dhaka, Bangladesh
Bachelor of Science in Computer Science — *2022 – 2026 (Expected)*
CGPA: 3.71 / 4.0 *(Merit Scholarship Recipient)* **Specialization:** Artificial Intelligence & Machine Learning
Relevant Coursework: Machine Learning, Data Science, AI, NLP, Pattern Recognition, Data Structures & Algorithms, Database Systems, Software Engineering.

Work Experience

United Commercial Bank PLC — Dhaka, Bangladesh
Intern — MD Secretariat, Process Reengineering (FinTech Young Talent Cohort) — *February 2026 – May 2026*
- One of 35 selected from 2,000+ applicants across Bangladesh for the bank's inaugural FinTech Young Talent Cohort; placed directly under MD Secretariat in the Process Reengineering vertical.
- Building a **Dashboard Manager** to replace manual Excel-based corporate task tracking for senior management, driving end-to-end process automation and digitization on the IT Audit team.

Eudika.com — Remote
Computer Science Instructor (Part-time) — *April 2025 – Present*
- Design and deliver O-Level Computer Science curriculum; mentor students in programming fundamentals and Cambridge exam technique.

MH Academy — Dhaka, Bangladesh
Instructor (Part-time) — *2022*
- Taught Mathematics, Physics, General Science, and IT to Grades 5–10 at an English-medium coaching center; adapted lesson delivery to diverse learning needs.

Technical Skills

- **Programming & Query Languages:** Python, SQL, C/C++, JavaScript, TypeScript, LaTeX
- **Data Analysis & ML:** Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, XGBoost, LightGBM, SHAP, LIME
- **Databases & Tools:** MySQL, PostgreSQL, Git, Linux, Jupyter Notebooks, Excel
- **Web Development:** React, Next.js, Flask, Node.js, TypeScript, Tailwind CSS

Projects

ZeroTex — LaTeX Resume Builder — *Next.js, TypeScript, React*
- Built production-ready resume builder with **Next.js 15, React 18, TypeScript, shadcn/ui**; engineered custom LaTeX generator with real-time preview, PDF export, dark/light mode. Deployed on **Vercel** at zerotex.vercel.app.

CampusCompanion — University Management Platform — *Next.js, TypeScript, Tailwind CSS*
- Developed campus platform with **Next.js, React, TypeScript, Tailwind CSS**; implemented SSR, authentication, course/event management, and announcements. Deployed at campuscompanion-flame.vercel.app.

CourseHUB — Course Management System — *Flask, Python, MySQL*
- Engineered course platform with **Flask, Python, MySQL**; built dual-dashboard with role-based authentication (Flask-Login, Flask-Bcrypt), course enrollment, resource management, and search functionality.

Red Light, Green Light — 3D Game — *Python, PyOpenGL*
- Created 3D FPS using **Python, PyOpenGL, GLUT**; engineered three-phase light system with freeze detection, enemy AI, combat mechanics with ammo, diverse enemy types, power-ups, and dynamic difficulty.

Mini-VSFS — Custom File System — *C, Systems Programming*
- Built custom file system in **C** with Superblock, Unix-style Inodes (128-byte), hierarchical directories (64-byte entries), bitmap allocation, CRC32 checksums, 12 direct pointers, and utilities for creating/adding files.

C Compiler — Intermediate Code Generator — *Flex/Yacc, C++*
- Built two-pass compiler using **Flex/Yacc, C++** with 16-node AST; implemented lexical/syntax/semantic analysis, hash-based symbol table, three-address code generation via post-order traversal, and optimized temp variable reuse.

Gamified Mobile Banking Platform — *React, Node.js, PostgreSQL*
- Built a full-stack web platform with **React, Node.js, PostgreSQL** to simulate gamified financial literacy; implemented portfolio simulation engine, user progress tracking, and an analytics dashboard measuring engagement and financial decision-making behaviour.

Research Experience

C-MAT: Cross-Modal Aligned Transformer for Neurodegenerative Assessment *B.Sc. Thesis*

- Leading development of **C-MAT**, a Dual-Stream Vision Transformer framework in **PyTorch** to diagnose Alzheimer's, Parkinson's, and FTD from multi-modal neuroimaging; converts EEG signals to spectrograms for ViT input and applies Cross-Modal Contrastive Alignment across MRI and EEG modalities.
- Implemented **Modality Dropout with Learnable Null Embeddings** for robust inference when only one modality is available (targeting low-resource settings); built an ETL pipeline aggregating **1,400+ records** from 5 international sources (BrainLat, OASIS, PPMI, AHEPA) with MNI152 registration and ICA-based EEG artifact rejection; integrated **SHAP** and Attention Rollout for brain region visualisation.

BioAlign-QLoRA: Biomedical Knowledge Graph Alignment *Research Project*

- Led team developing novel **Knowledge Graph Separation** metric; analyzed 68,444+ gene-disease associations using **Python (Pandas, NumPy)** and fine-tuned three LLMs (Llama-3, Mistral, Phi-3) with **QLoRA, PyTorch, Hugging Face**.
- Achieved **83.8% accuracy**, outperforming BioMistral-7B; co-authored research paper demonstrating **126% improvement** in embedding alignment, preparing for journal submission.

Breast Cancer Prediction via Multi-Task U-Net *Deep Learning*

- Preprocessed BUSI dataset (600+ ultrasound images) using **Python, PyTorch, OpenCV, Albumentations**; architected multi-task U-Net for simultaneous segmentation and classification (benign/malignant/normal).
- Achieved **85.4% Dice coefficient, 87.5% accuracy**; conducted statistical validation using **Scikit-learn** and documented methodology in research report with visualizations.

Recession Prediction with Explainable AI *Research Project*

- Integrated economic data from 15+ sources using **Python (Pandas, NumPy), SQL**; engineered lagged features and rolling statistics to predict GDP-Based and Sahm Rule indicators.
- Benchmarked models (Random Forest, XGBoost, LSTM, Ensemble) achieving **96% accuracy, 0.97 weighted F1-score**; applied **SHAP** for interpretability; authored research paper with visualizations.

Exoplanet Habitability Classification *ML Research*

- Built ML pipeline analyzing PHL Catalog (5,600+ exoplanets, 100+ features) using **Python (Pandas, Dask, Matplotlib, Seaborn)**; conducted EDA, handled missing data, and benchmarked 8 algorithms.
- Addressed class imbalance with SMOTE; implemented **LIME** for explainability and permutation importance; documented complete workflow in research report with visualizations.

Extracurricular Activities

Volunteer | Nishonkoch Foundation 2018 – 2020

- Contributed to community service initiatives; assisted in developing social media fundraising campaigns and coordinating outreach events.

Sub Executive, Marketing & PR | BRAC University Business Club 2022 – 2024

- Executed multi-platform digital marketing campaigns promoting university events; tracked campaign metrics and prepared performance reports.

Senior Executive, Business Development | IABC | BRACU 2022 – 2023

- Organized Model UN events, led club promotion initiatives, and managed member recruitment drives.

Awards & Certifications

Silver Explorer Award — 3rd International Poster Competition for Young Researchers, UNIV (2025);
awarded for research on gamified mobile banking adoption among university students.

Outstanding Delegate — BUPIMUN (2023); **Special Mention** — NDCMUN (2020), EMUNGA (2021)

Certifications:

- AWS Educate Machine Learning Foundations (Amazon Web Services, 2025)
- Introducing Generative AI with AWS (Udacity, 2025)
- Exploratory Data Analysis in Python & SQL (DataCamp, 2024)

Languages

English: Fluent (IELTS 7.5) **Bangla:** Native **Hindi / Urdu:** Conversational